

NAMAN SHARMA

Robotics & Computer Vision Engineer | ROS 2 | Edge AI | Simulation
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naman2328.github.io/naman-robotics-vision-portfolio

PROFILE

Final-year Mechatronics Engineer with 12 months of hands-on production deployment at a Tier-1 automotive supplier. Deep practical fluency in ROS 2, computer vision, and edge AI with real robots and real deadlines. Bias for clean architecture and reliable real-world performance.

EXPERIENCE

Process Engineering Intern — Autoliv India

Tier-1 Automotive Supplier · ISO-Certified · Chandigarh · Feb 2025 – Feb 2026

- Designed and deployed a vision-based OCR inspection system on NVIDIA Jetson Orin Nano Super — ~95% accuracy at 150–250 ms latency per frame under live factory conditions.
- Engineered an industrial thread-detection pipeline (color-based ROI classification); reduced manual inspection workload by ~40%.
- Optimised real-time inference pipelines for sustained throughput within strict edge power and thermal budgets — no GPU server, no cloud dependency.
- Delivered an operator-facing OK/NG GUI with decision logging; adopted by QC team with zero retraining required.
- Collaborated across mechanical, quality, and process teams in an ISO-compliant environment; translated output into auditable documentation.

PROJECTS

Vision-Guided Pick & Place — ROS 2 + Dobot Magician

[ROS 2](#) · [OpenCV](#) · [Python](#) · [Real Hardware](#)

- Decoupled vision and control via separate ROS 2 nodes; real-time object localisation and pose estimation using OpenCV.
- 2D camera-to-robot coordinate transformation for precise grasping; achieved ± 2 mm positional repeatability.
- Eliminated servo jitter through PID gain tuning; clean pub/sub architecture separating perception, planning, and execution.
- Deployed on physical Dobot hardware — modular design allows vision pipeline to be swapped independently.

Industrial OCR Inspection System — Jetson Edge Deployment

[PaddleOCR](#) · [OpenCV](#) · [Python](#) · [Jetson Orin Nano](#) · [Qt](#)

- ROI-based pipeline with confidence thresholding for deterministic OK/NG decisions under factory conditions (glare, inconsistent positioning).
- Full offline deployment on Jetson Orin Nano Super — no cloud or GPU server dependency.
- Qt-based production UI with real industrial camera integration; result logging for audit trail and traceability.

Omnidirectional Mecanum Mobile Robot

[Python](#) · [Motor Control](#) · [Mecanum Kinematics](#) · [ROS 2](#) · [Embedded Systems](#)

- Holonomic mobile robot using Mecanum wheels — full kinematic model for arbitrary 2D motion without chassis rotation.
- Validated lateral translation, diagonal motion, and in-place rotation; ROS 2-ready platform for SLAM and navigation integration.
- Per-wheel velocity vector control for precise omnidirectional movement.

Custom 3D-Printed Multi-DOF Robotic Arm

[Python](#) · [Servo Motors](#) · [3D Printing](#) · [Mechanical Design](#) · [PWM Calibration](#)

- Designed, 3D-printed, and assembled a multi-DOF arm from scratch — full hardware-software integration loop.
- PWM-calibrated servo control for precise joint positioning; motion sequences for repeatable trajectories.
- Diagnosed and resolved real-world backlash and torque limit constraints absent from simulation.

TECHNICAL SKILLS

Robotics: ROS 2 (Humble) · SLAM · Sensor Fusion · Motion Planning · URDF · Robot Kinematics · LiDAR · Encoders

Simulation: Gazebo (Fortress / Classic) · Isaac Sim · RViz · URDF/Xacro

Computer Vision: OpenCV · PaddleOCR · ROI Pipelines · Object Detection · Image Preprocessing · OK/NG Logic

Edge AI & Deploy: NVIDIA Jetson Orin Nano Super · TensorRT · ONNX · Real-time Inference Optimisation

Languages: Python (primary) · C++ (working) · Bash

Hardware: Jetson Orin Nano · Raspberry Pi · Arduino · Dobot Magician · Mecanum Drive · Servo/Motor Control

Tools: Git · Linux (Ubuntu) · Qt · ISO-compliant documentation · Deterministic system design

EDUCATION

B.E. Mechatronics Engineering — Chitkara University, Punjab

2022 – 2026 · Relevant coursework: Robotics · Control Systems · Embedded Systems · Machine Vision · Mechatronics System Design